Md Hashibul Hossain

Junior Research Scientist, CredoSense (Dhaka) · MSc (Thesis) in Forestry, With Distinction

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Research profile

Forestry-trained environmental scientist working on how ecosystems respond to environmental stress through two complementary paths. In the wet lab I study fungal heavy-metal tolerance and bioremediation, plant pigments and nutrition, and plant pathogens, using culture assays, antioxidant enzyme profiling and ICP-OES. In the dry lab I use satellite remote sensing, drone photogrammetry and computer vision to map vegetation, land cover and contamination risk, and I write reproducible R and Python analyses. MSc (Thesis) in Forestry With Distinction; currently Junior Research Scientist at CredoSense, Dhaka. Seeking a PhD that links lab-measured mechanisms with landscape-scale environmental and ecological questions.

Research interests

Environmental microbiology · fungi–heavy metal interactions and mycoremediation · plant pathology and pathogen detection · plant physiology and nutrient chemistry · mycorrhizal ecology and soil carbon · remote sensing of vegetation and land cover · drone photogrammetry · computer vision for biology · urban green space and landscape ecology · fungal–plant–animal interactions in tropical forest restoration

Education

Session 2022 – 2024

MSc (Thesis) in Forestry

Shahjalal University of Science & Technology (SUST), Sylhet

CGPA 3.80 / 4.00 · With Distinction · result published Feb 2026

- Classes began Mar 2024; thesis defended Jan 2026 (schedule delayed by the 2024 campus closures)
- Thesis: tolerance, bioaccumulation and bioremediation of heavy metals in water and soil by Aspergillus niger and Trichoderma asperellum, with screening of metallo-tolerant landfill isolates
- Supervisor: Dr Mohammed Masum Ul Haque · Co-supervisor: Prof. Romel Ahmed
- Data Analysis and Visualization in R (A+); Dissertation and Thesis both A+

Session 2018 – 2022

BSc (Honours) in Forestry

Shahjalal University of Science & Technology (SUST), Sylhet

CGPA 3.43 / 4.00 · result published May 2024

- Final examinations held Dec 2023 (schedule delayed by COVID-19)
- Final-year project (2022 – 2023): in-vitro assessment of heavy-metal tolerance of selected fungi
- Coursework: Remote Sensing and GIS, Mycology and Forest Pathology, Genetics and Tree Improvement, Research Methodology

Research experience

Jul 2026 – present

Junior Research Scientist

CredoSense (Canada-based) · Dhaka office, Bangladesh

- Machine-learning and computer-vision models that detect pathogenic fungi from microscopic spore images
- Object-detection models for greenhouse pests: curation, annotation and evaluation
- Plant pigment extraction (chlorophyll, carotenoids) and spectrophotometry
- Plant-tissue digestion and macro/micronutrient analysis by ICP-OES

Jan – Dec 2025

Research Associate

Dept. of Forestry & Environmental Science, SUST

- Designed and led a fungal bioremediation project, including landfill sampling and isolation
- Isolated fungi from resin-forming agarwood trees
- Operated ICP-OES independently; UV-Vis colorimetric and enzyme assays
- Statistics in R; presented at international conferences; mentored junior researchers

Jan – Dec 2024

Research Assistant

Dept. of Forestry & Environmental Science, SUST

- Experiment design for plant–pathogen interaction and metal-stress studies
- Ran seed-susceptibility tests against fungi
- Literature review, laboratory protocols, controlled sample handling

Jan 2024

Intern and Field Researcher

USAID Bangladesh Ecosystems / Protibesh Activity · Khadimnagar National Park · Supervisor: Md. Modinul Ahsan, PhD

- Compulsory internship in protected-area biodiversity management
- Mixed-methods study of co-management office bearers' and People's Forum members' knowledge of the Protected Area Management Rules, 2017
- 22 respondents from 10 villages (purposive sampling, structured questionnaire, field observation); analysis in R 4.3

Fieldwork

11 sites

Agarwood plantation survey

Surveyed 11 plantations alone in about 10 days; soil sampling and tree inventory in plots covering 5% of plantation area

Camera traps

Camera-trap study

Chose camera-trap sites for nocturnal carnivores from food-resource distribution; mapped and analysed detections in GIS

Landfill

Landfill rhizosphere sampling

GIS-selected site; 7 root-zone soil and 3 water samples; aseptic collection, cold transport, –80 °C storage

Forest types

Forest types worked in

Hill forest (Sylhet, Chittagong Hill Tracts), sal forest (Madhupur), mangrove (Sundarbans)

Selected research projects

2024 – 2026

Heavy-metal tolerance, bioaccumulation and bioremediation by filamentous fungi

MSc thesis · SUST · Supervisors: Dr M. M. U. Haque, Prof. R. Ahmed

Dose–response screening of Aspergillus niger and Trichoderma asperellum against As(V), Cd(II), Cr(VI) and Pb(II), with enzyme profiling and removal tests in water and soil microcosms.

2024 – 2025

Metallo-tolerant fungi from the Lalmatia waste dump

MSc field discovery pipeline · Sylhet

GIS-chosen sampling site, 10 field samples (7 soil, 3 water), bioavailable metals by ICP-OES, and isolation of 27 morphotypes by serial dilution.

- ITS-based identification in progress
- Factorial assays of up to 210 plates in one day (7 fungi × 2 metals × 5 concentrations × 3 replicates)
- Prepared and counted spore suspensions to inoculate soil in an ongoing study of the 7 isolates

2025 – present

Fungal associations of resin-forming agarwood

Research Associate · SUST

Isolation of fungi from resin-forming Aquilaria trees to study fungus–plant associations.

- Isolated fungi from resinous wood and helped inoculate living agarwood trees (results pending)

2024

Mycorrhizal fungi of dipterocarp forest trees

Culture work · SUST

Cultured fungi from fruit bodies associated with telsur (*Hopea odorata*) and garjan (*Dipterocarpus*) on MMN medium.

2024 – 2025

Multi-sensor GIS site selection for contamination sampling

MSc thesis · ArcMap 10.8 · Google Earth Engine · R

Multi-criteria suitability model fusing atmospheric, thermal, vegetation, terrain and hydrology layers on a common 500 m grid over a 10-year window, with sensitivity analysis.

2022 – 2023

In-vitro heavy-metal tolerance of three fungi

BSc project · SUST

A. niger, R. stolonifer and T. asperellum on PDA amended with Pb, Hg and As at 100–500 ppm, growth measured at 24, 48 and 72 h.

Dec 2024 – Feb 2025

YOLO11s leaf detector with TensorFlow.js export

Python & Data Science training · IICT-SUST

Transfer-learned YOLO11s for 2-class leaf detection (1,107 train / 46 val images, 100 epochs), validated and exported for in-browser inference.

2023 **Land-cover change, Karnaphuli Range (Kaptai)**
Coursework · SUST
Satellite image analysis of land-use and land-cover change across a ~560 ha forest range.

Research software & open code

Python
AGPL-3.0
Dīvi: drone mapping and canopy analysis for researchers
github.com/Mraz11/Divi
Offline-first toolkit for consumer RGB drones (formerly Avispria): terrain-aware DJI Fly mission planning, photo QA, WebODM orthophoto/DSM processing, canopy height, cover, gaps and tree-crown analysis. 17.9k lines of Python, 1,053 automated tests. Early version; one test flight (DJI Lito X1, 79 photos), results not yet validated.

R
MIT
Estimating stress tolerance: dose–response modelling done properly
github.com/Mraz11/dose-response-ec50
LL.4 reproduces the published ryegrass ED50 exactly (3.058, 95% CI 2.671–3.445). On labelled synthetic data, RMSE of log-EC50 falls from 0.21 to 0.055 across 12 designs, 1,000 simulations each.

R
MIT
Natural background vs anthropogenic enrichment of toxic elements in US soils
github.com/Mraz11/soil-metal-enrichment
Zn, Cd and Ni show land-cover-linked enrichment robust to the normaliser. Pb's strongest Al-based signal (p = 6.2e-13) did not replicate with Ti (p = 0.68) and is reported as unconfirmed.

R
MIT
Climate and ecosystem controls on global soil respiration
github.com/Mraz11/climate-controls-soil-respiration
MAT × biome interaction is supported by LRT (p = 0.036; 0.0019 after removing 28 influential points) but does not improve prediction: exact leave-one-study-out CV over 1,247 studies (2,494 refits) gives predictive $R^2 \approx 0.11$.

Python
MIT
Metal-resistance genes in bacteria from contaminated vs reference environments
github.com/Mraz11/metal-resistance-genes-environment
Contaminated genomes carry about 4% more metal-resistance genes per 1,000 proteins (rate ratio 1.039, 95% CI 0.996–1.083). Significant with HC1-robust errors (p = 0.030) but not once clustered by sampling site (p = 0.119) or clade (p = 0.076): 84 genomes cannot confirm the effect.

R
MIT
Dose–time metallotolerance screening in fungi
github.com/Mraz11/dose-time-metallotolerance-fungi
R pipeline producing Figures 1–8 for A. niger, T. asperellum and R. stolonifer under As, Pb and Hg: tolerance index, growth rate, Tukey contrasts, TI heatmap, LL.4 curves, IC50 with bootstrap CI and censoring.

Publications & presentations

MANUSCRIPTS IN REVIEW

1. **Hossain, M.H.**, Haque, M.M.U., Ahmed, R., & Rahman, M.A. *Integrated screening of metal tolerance and short-contact dry-biomass biosorption reveals metal-specific performance in filamentous fungi.* In review
2. Aziz, F.B.A., Sultana, R., Masum, K.M., & **Hossain, M.H.** *Potential urban green space suitability appraisal using GIS-based multi-criteria analysis: A case study of North-eastern Sylhet City, Bangladesh.* In review
3. Rekha, P.R., Ghose, A., **Hossain, M.H.**, & Masum, K.M. *Urban green spaces as carnivore refugia: A camera-trap study in a biodiversity-rich university campus in Bangladesh.* In review

CONFERENCE PAPER

4. **Hossain, M.H.**, Ahmed, R., & Haque, M.M.U. (2025). *Heavy metal tolerance by fungi: A step toward mycoremediation strategy for contaminated soils.* International Conference and Bioscience Carnival, Rangamati Science and Technology University. doi:10.13140/RG.2.2.29154.03525

POSTERS

5. **Hossain, M.H.** (2023). *Mycorrhizal role of carbon sequestration in forest soil.* Contemporary Trends in Forestry Research seminar, SUST.
6. **Hossain, M.H.** (2023). *Bioaccumulation of heavy metals by fungi and its possible effects on the environment.* International Day of Forests, SUST.

REPORTS (NON-PEER-REVIEWED)

7. **Hossain, M.H.**, & Eva, N.J. (2024). *Knowledge of office bearers and People's Forum members of Khadimnagar National Park about the Protected Area Management Rules, 2017.* BSc internship report, USAID Ecosystems / Protibesh Activity.
8. **Hossain, M.H.** (2023). *Image analysis of Karnaphuli Range.* Coursework report, SUST.

Skills

Wet lab & analytical chemistry
ICP-OES · Microwave digestion · UV-Vis · Pigment extraction · CAT · SOD · GR assays · Aseptic culture · Serial dilution · Cryopreservation · Bioremediation microcosms · Plant pathology

Molecular
DNA extraction (fungi, bacteria) · PCR: ITS and 18S markers · BLAST · Phylogenetic analysis · FastQC on NCBI sequence data · ITS identification of DMFT1–7 (in progress) · Comparative genomics: DIAMOND, Mash

Statistics & code
R · Python · pandas · ggplot2 / seaborn · LL.4 dose–response · Bootstrap CI · Mixed models (lme4) · Cross-validation · ANOVA / Tukey · PCA · NMDS · renv · uv · Make · pytest · Git / GitHub

GIS & remote sensing
Google Earth Engine · ArcMap 10.8 · MCDA · Drone photogrammetry (WebODM) · DJI WPML · Sentinel-2 / 5P · Landsat LST · MAIAC AOD · VIIRS · SRTM · Land-cover change · i-Tree

Machine learning & vision
YOLO11 / Ultralytics · Transfer learning · Roboflow · Annotation · TensorFlow.js export · ImageJ · Microscopy image analysis

Field & social-science methods
Field sampling · Structured questionnaires · Purposive sampling · Stakeholder interviews · Mixed-methods analysis · Protected-area co-management

Training & certification

Jun 2025	**WEkEO for Climate Change** · Copernicus & Mercator Ocean International
Apr 2025	**Earth Observation for Climate Adaptation and Resilience in South Asia** · AIRCAS (CAS), Jagannath University, BRAC University C3ER
Feb 2025	**Core Techniques in Molecular Biology and NGS (5 days)** · Central Dogma Lab, Invent Technologies
Dec 2024 – Feb 2025	**Python Programming and Basic Data Science (80 h)** · IICT-SUST / Bangladesh Computer Council
Feb 2025	**Understanding Climate Change and Climate Finance** · PKSF / Green Climate Fund
Sep 2024	**ICP-OES (8 days)** · Invent Technologies at SUST
Sep 2023	**Bioinformatics Bootcamp 2.0** · CHIRAL Bangladesh
Jul 2023	**Urban Tree Inventory using i-Tree** · USFS International Programs

Leadership & service

Mar 2023 – Mar 2024	**G Studio Wing Chief** · Green Explore Society, SUST (9th Executive Committee)
Mar 2022 – Mar 2023	**Assistant Publication Secretary** · Green Explore Society, SUST (8th Executive Committee)
2025	**Mentor to junior researchers** · Research Associate role, Dept. of Forestry & Environmental Science, SUST

Awards & honours

2026	MSc awarded With Distinction
2025	National Eco Olympiad — National Round
2018	Bangladesh Physics Olympiad — Regional round
2017 – 2018	Bangladesh Chemistry Olympiad — Preliminary rounds (7th, 8th)
2017	Bangladesh Mathematical Olympiad — Regional round
2017	Bangladesh Botany Olympiad — Participant

Languages

English	IELTS Academic 7.5 overall (CEFR C1), Sep 2026 · Listening 8.0, Reading 8.5, Writing 6.5, Speaking 6.5
Bengali	Native

Referees

Dr Mohammed Masum Ul Haque
Associate Professor, Forestry & Environmental Science, SUST
BSc and MSc supervisor

Prof. Romel Ahmed
Professor, Forestry & Environmental Science, SUST
MSc co-supervisor

Prof. Kazi Mohammad Masum
Professor, Forestry & Environmental Science, SUST
Collaborator on the camera-trap manuscript and another ongoing paper

Contact details, and an additional referee from CredoSense, available on request.